Virtual-gate matrix — Test split
4-6-8-12-16-24_rays_40-60-100_points_500_samples

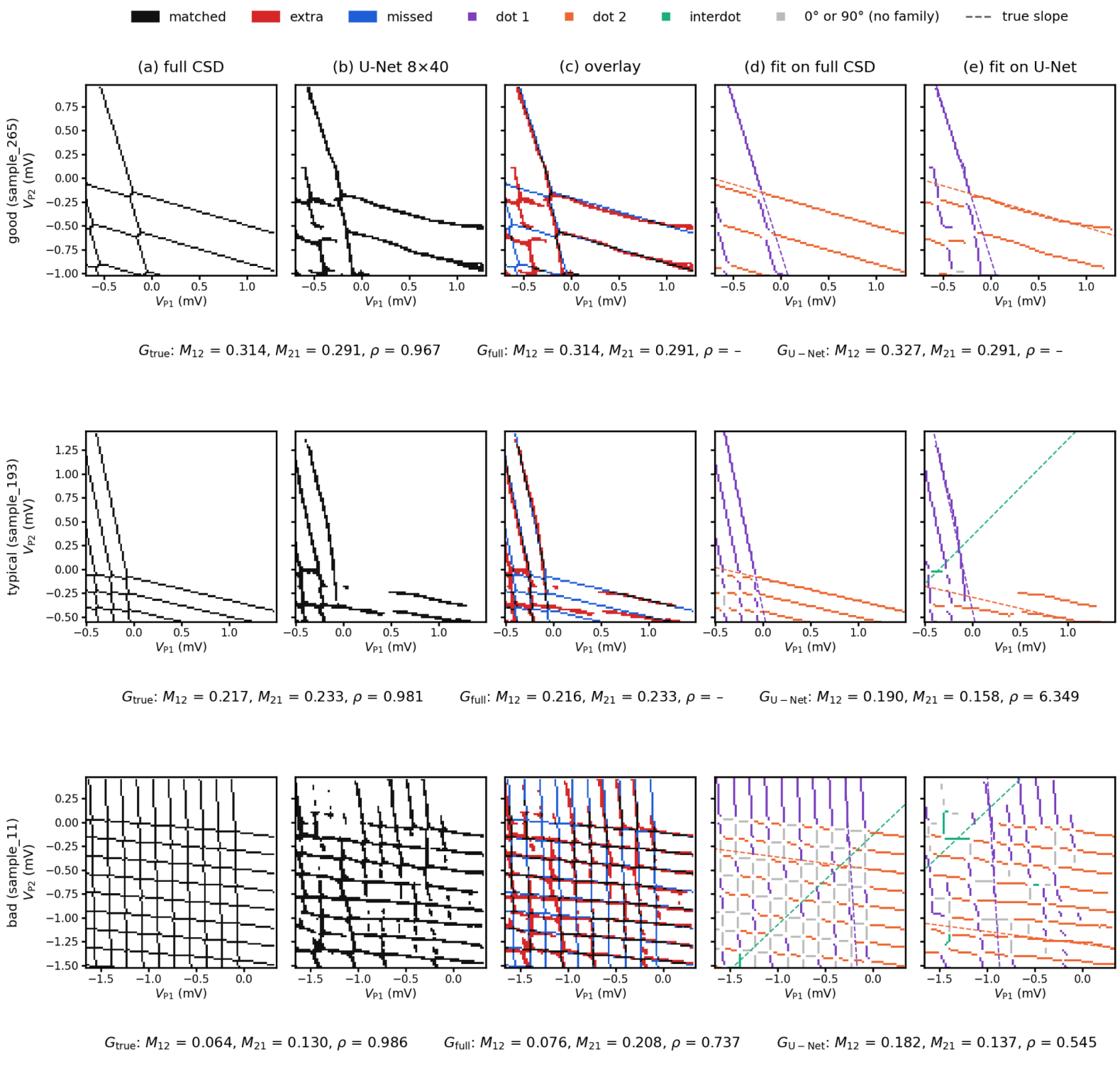
method	rays × points	coverage %	dθ1 (°)	dθ2 (°)	dθ_int (°)	M12 rel. err. (%)	M21 rel. err. (%)	rho rel. err. (%)	interdot found	failed
full CSD	—	100.00	0.09 ± 0.26 [0.06]	-0.02 ± 0.86 [0.04]	-10.86 ± 31.16 [28.27]	1.15 ± 3.40 [0.53]	1.21 ± 8.72 [0.36]	270.56 ± 369.65 [39.58]	8/50	0/50
U-Net	4 × 40	1.57	-3.36 ± 7.93 [2.90]	2.33 ± 7.69 [3.20]	-17.32 ± 31.87 [35.26]	-2.28 ± 49.74 [29.30]	1.49 ± 49.45 [28.53]	156.65 ± 192.20 [133.75]	36/50	0/50
U-Net	4 × 60	2.35	-2.73 ± 7.84 [3.85]	2.16 ± 7.72 [2.35]	-11.51 ± 32.93 [36.48]	2.12 ± 49.39 [30.65]	6.27 ± 62.80 [19.95]	126.89 ± 180.71 [75.07]	39/50	0/50
U-Net	6 × 40	2.33	-2.68 ± 7.51 [2.40]	2.02 ± 5.98 [2.62]	-5.67 ± 34.42 [34.62]	-0.72 ± 47.22 [23.42]	2.06 ± 41.95 [23.60]	109.79 ± 193.20 [71.54]	27/50	0/50
U-Net	8 × 40	3.11	-2.77 ± 6.81 [1.82]	2.30 ± 6.51 [1.88]	-6.47 ± 31.41 [32.55]	-3.66 ± 40.08 [15.82]	-1.89 ± 33.55 [12.90]	113.13 ± 192.39 [64.81]	22/50	0/50
U-Net	6 × 60	3.52	-1.01 ± 5.40 [2.57]	0.42 ± 7.30 [2.23]	-19.46 ± 30.55 [36.88]	7.87 ± 48.37 [21.59]	11.85 ± 59.13 [21.46]	170.13 ± 201.22 [140.84]	32/50	0/50
U-Net	12 × 40	4.59	-0.01 ± 3.89 [1.33]	1.33 ± 5.18 [1.75]	-22.56 ± 27.54 [35.00]	9.44 ± 41.39 [10.87]	1.52 ± 31.36 [13.79]	212.28 ± 210.25 [174.63]	19/50	0/50
U-Net	16 × 40	6.08	-1.41 ± 4.09 [1.37]	0.26 ± 6.27 [1.36]	-13.80 ± 33.48 [37.65]	-0.94 ± 30.18 [11.00]	6.75 ± 50.85 [13.55]	156.34 ± 213.60 [81.45]	15/50	0/50
U-Net	24 × 40	8.89	0.58 ± 3.97 [1.08]	-0.33 ± 3.03 [1.24]	-8.11 ± 24.66 [19.60]	12.40 ± 42.29 [10.39]	6.80 ± 21.56 [10.41]	74.75 ± 150.22 [39.14]	12/50	0/50

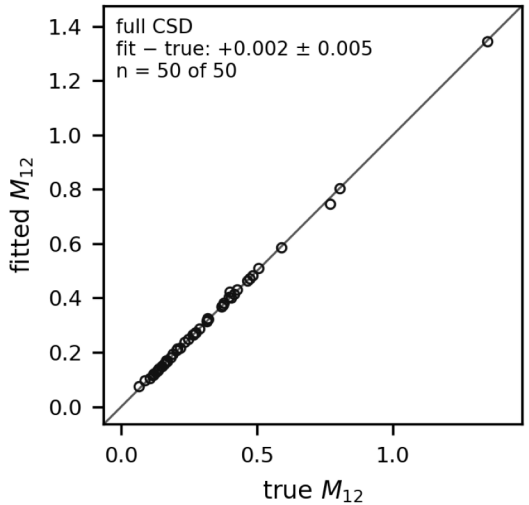
Each error cell: mean ± std over devices where the family was found, median |error| in brackets. 'failed': devices with no dot-1 or no dot-2 piece. rho only where an interdot family was found. The U-Nets were given 1.57–8.89 % of the grid (4 × 40 to 24 × 40 rays × points). On the full CSD the same line fit recovers M12 and M21 with a median relative error of 0.53 % and 0.36 % (41/50 and 44/50 devices within 0.005 of the true value): the floor of the fit itself on 100 × 100 px maps. From the U-Net maps the median relative error of M12 falls from 29.3 % at 1.57 % coverage to 10.4 % at 8.89 %, and that of M21 from 28.5 % to 10.4 %; at 8 × 40 (3.11 %) it is 15.8 % and 12.9 %. The U-Net error is not uniform: at 8 × 40 the devices with a true coupling above 0.4 are underestimated (median -46 % for M12, n = 11; -36 % for M21, n = 10), while those below 0.4 are nearly unbiased (+2.3 % and +0.5 %). The interdot slope, and so rho, is not recovered: interdot segments are a few pixels long, the full CSD yields an interdot family on only 8/50 devices (median |dθ_int| 28.3°, median |rho error| 40 %), and the 12–39 devices per U-Net on which an interdot family is found have median |dθ_int| of 20–38°, i.e. spurious pieces. rho is therefore reported but not claimed. A dot family was missing on 0/50 devices for the full CSD and on at most 0/50 for any U-Net.

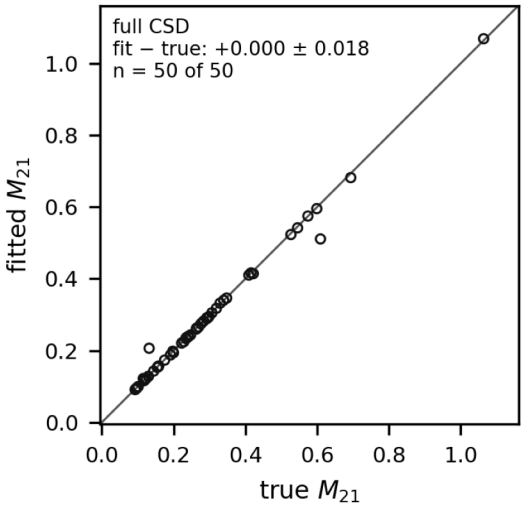
Virtual-gate matrix — Validation split (chose the thresholds; not held out)
4-6-8-12-16-24_rays_40-60-100_points_500_samples

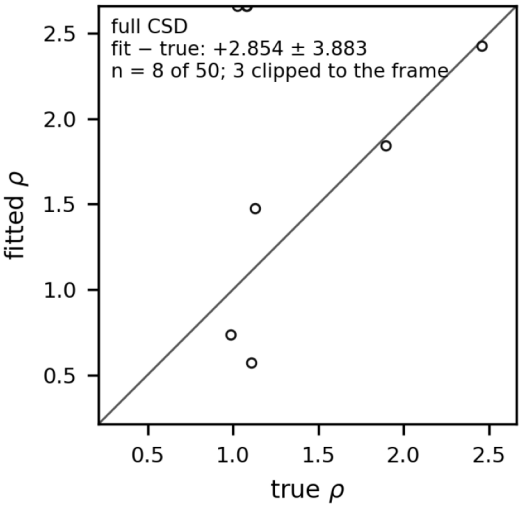
method	rays × points	coverage %	dθ1 (°)	dθ2 (°)	dθ_int (°)	M12 rel. err. (%)	M21 rel. err. (%)	rho rel. err. (%)	interdot found	failed
full CSD	—	100.00	0.10 ± 0.76 [0.05]	-0.10 ± 0.68 [0.04]	-5.56 ± 16.08 [4.07]	1.11 ± 7.03 [0.30]	1.11 ± 7.26 [0.31]	80.21 ± 192.48 [9.52]	16/75	0/75
U-Net	4 × 40	1.57	-8.29 ± 12.39 [4.24]	4.92 ± 12.09 [3.05]	-14.81 ± 33.99 [38.27]	-21.13 ± 41.91 [31.50]	-0.44 ± 55.47 [29.05]	154.48 ± 202.84 [125.38]	56/75	0/75
U-Net	4 × 60	2.35	-7.08 ± 11.47 [2.91]	4.50 ± 11.40 [3.12]	-11.87 ± 29.72 [30.73]	-16.93 ± 41.49 [26.85]	-0.36 ± 54.47 [26.75]	101.05 ± 155.72 [71.31]	58/75	0/75
U-Net	6 × 40	2.33	-4.98 ± 9.55 [2.13]	3.85 ± 10.37 [2.31]	-11.56 ± 28.65 [29.50]	-12.50 ± 32.53 [19.55]	0.73 ± 49.03 [19.33]	113.17 ± 194.35 [72.88]	44/75	0/75
U-Net	8 × 40	3.11	-4.58 ± 10.20 [2.28]	3.45 ± 9.52 [2.13]	-5.83 ± 29.39 [25.88]	-10.06 ± 32.71 [16.19]	-3.22 ± 39.00 [14.47]	74.43 ± 151.64 [61.60]	40/75	0/75
U-Net	6 × 60	3.52	-4.69 ± 10.70 [2.83]	3.22 ± 9.93 [1.93]	-17.78 ± 27.62 [32.12]	-7.33 ± 39.47 [21.90]	0.57 ± 44.67 [17.05]	149.08 ± 202.16 [80.55]	39/75	0/75
U-Net	12 × 40	4.59	-3.51 ± 9.18 [1.90]	2.62 ± 10.08 [1.29]	-12.47 ± 25.67 [25.95]	-5.81 ± 33.51 [13.04]	2.09 ± 40.07 [9.91]	93.59 ± 154.22 [43.49]	18/75	0/75
U-Net	16 × 40	6.08	-3.53 ± 9.06 [1.66]	2.79 ± 8.95 [1.19]	-19.01 ± 31.51 [36.92]	-6.96 ± 29.99 [10.66]	-2.89 ± 35.09 [10.16]	188.19 ± 255.43 [131.97]	21/75	0/75
U-Net	24 × 40	8.89	-1.94 ± 7.11 [0.66]	1.29 ± 7.24 [0.71]	-7.58 ± 30.13 [26.12]	-0.93 ± 27.35 [5.60]	2.52 ± 31.61 [5.94]	109.58 ± 244.71 [51.06]	19/75	0/75

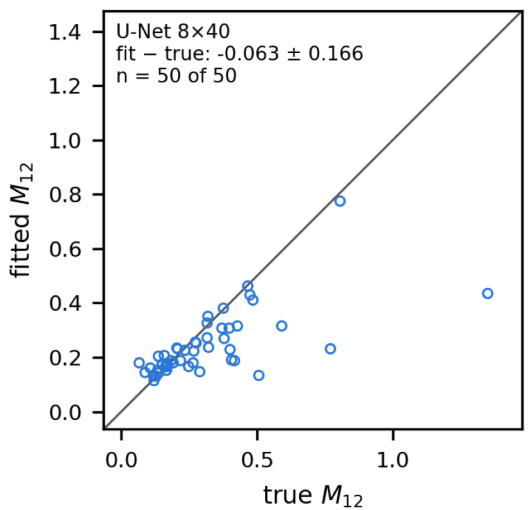
Each error cell: mean ± std over devices where the family was found, median |error| in brackets. 'failed': devices with no dot-1 or no dot-2 piece. rho only where an interdot family was found.

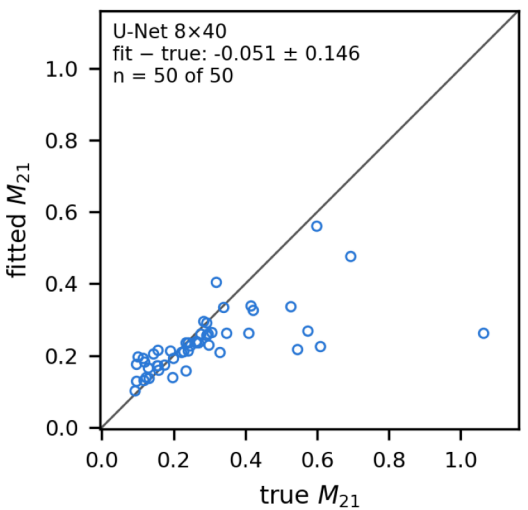

matched
extra
missed
dot 1
dot 2
interdot
0° or 90° (no family)
true slope
(a) full CSD
(b) U-Net 8×40
(c) overlay
(d) fit on full CSD
(e) fit on U-Net
good (sample_265)
typical (sample_193)
bad (sample_11)
V_{P2} (mV)
V_{P1} (mV)
G_{true}: M_{12} = 0.314, M_{21} = 0.291, ρ = 0.967
G_{full}: M_{12} = 0.314, M_{21} = 0.291, ρ = –
G_{U-Net}: M_{12} = 0.327, M_{21} = 0.291, ρ = –
G_{true}: M_{12} = 0.217, M_{21} = 0.233, ρ = 0.981
G_{full}: M_{12} = 0.216, M_{21} = 0.233, ρ = –
G_{U-Net}: M_{12} = 0.190, M_{21} = 0.158, ρ = 6.349
G_{true}: M_{12} = 0.064, M_{21} = 0.130, ρ = 0.986
G_{full}: M_{12} = 0.076, M_{21} = 0.208, ρ = 0.737
G_{U-Net}: M_{12} = 0.182, M_{21} = 0.137, ρ = 0.545

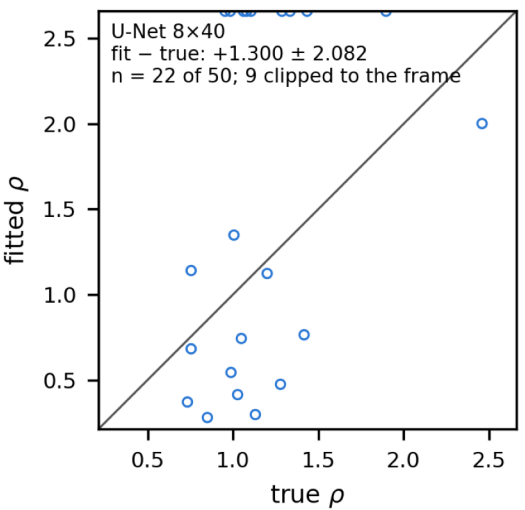

full CSD
fit − true: +0.002 ± 0.005
n = 50 of 50
fitted M_{12}
true M_{12}

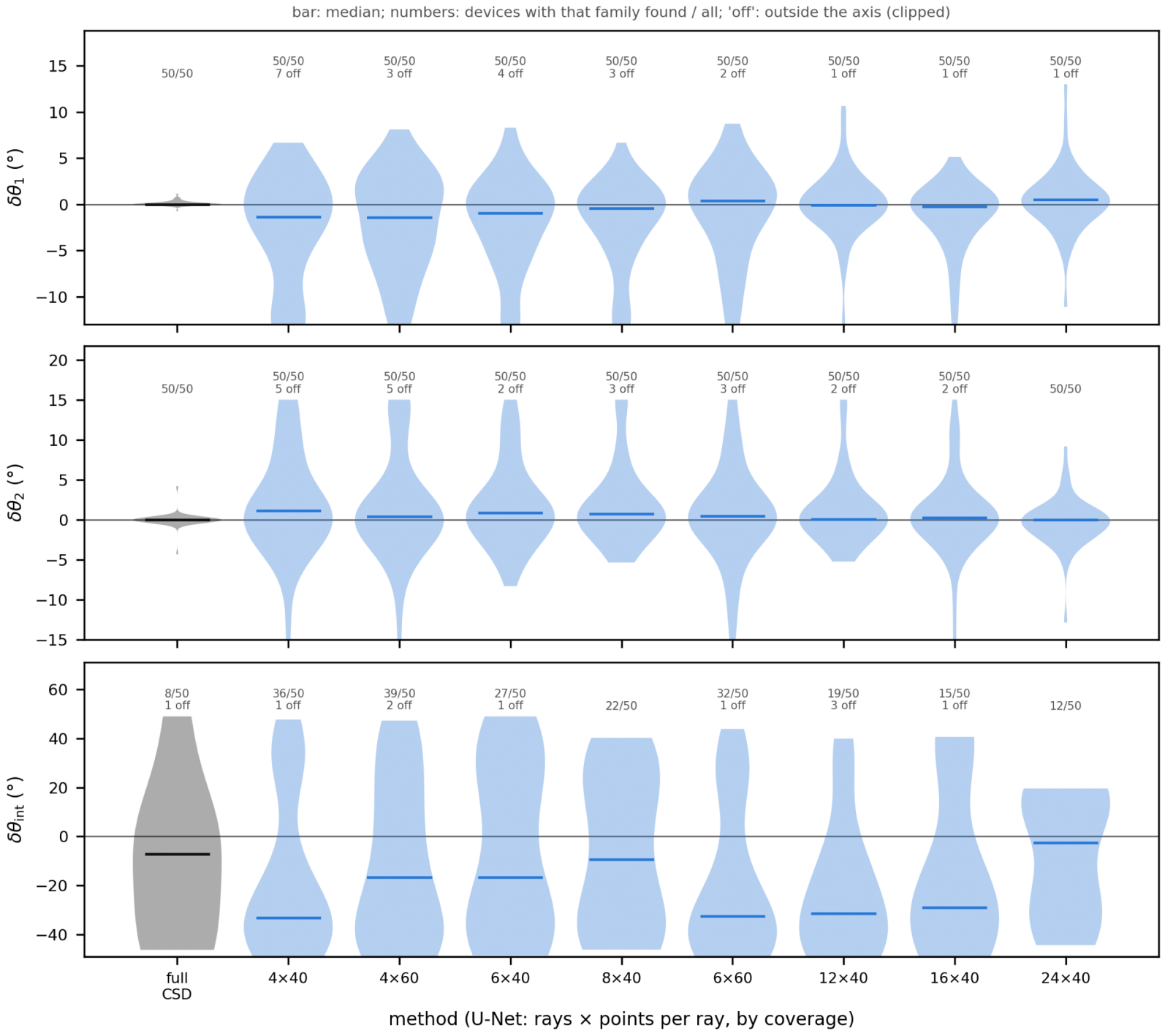

full CSD
fit − true: +0.000 ± 0.018
n = 50 of 50
fitted M_{21}
true M_{21}

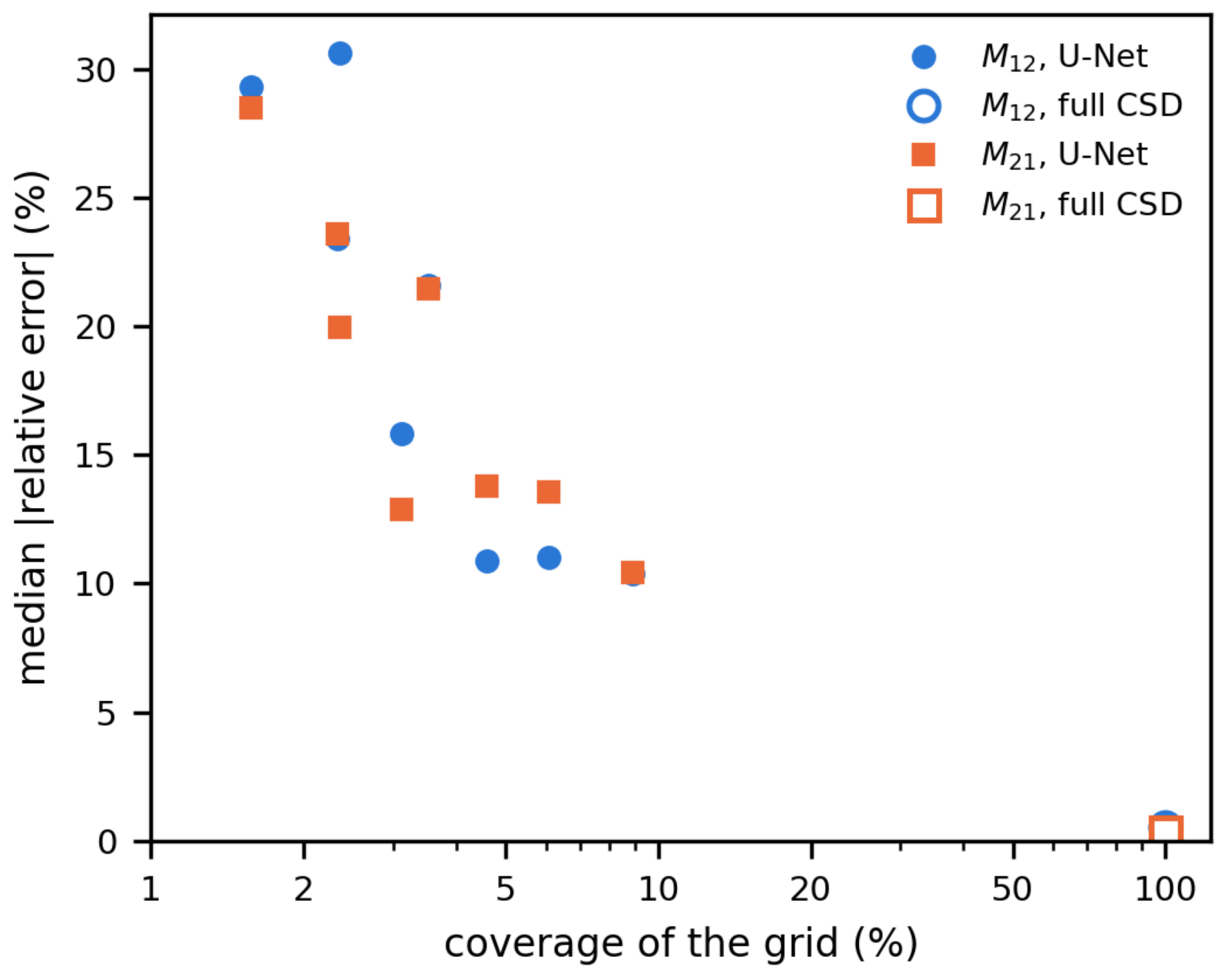

full CSD
fit − true: +2.854 ± 3.883
n = 8 of 50; 3 clipped to the frame
fitted ρ
true ρ


U-Net 8×40
fit − true: -0.063 ± 0.166
n = 50 of 50
fitted M_{12}
true M_{12}


U-Net 8×40
fit − true: -0.051 ± 0.146
n = 50 of 50
fitted M_{21}
true M_{21}


U-Net 8×40
fit − true: +1.300 ± 2.082
n = 22 of 50; 9 clipped to the frame
fitted ρ
true ρ


bar: median; numbers: devices with that family found / all; 'off': outside the axis (clipped)
$\delta\theta_1$ (°)
50/50
50/50
7 off
50/50
3 off
50/50
4 off
50/50
3 off
50/50
2 off
50/50
1 off
50/50
1 off
50/50
1 off
$\delta\theta_2$ (°)
50/50
50/50
5 off
50/50
5 off
50/50
2 off
50/50
3 off
50/50
3 off
50/50
2 off
50/50
2 off
50/50
$\delta\theta_{int}$ (°)
8/50
1 off
36/50
1 off
39/50
2 off
27/50
1 off
22/50
32/50
1 off
19/50
3 off
15/50
1 off
12/50
full CSD
4×40
4×60
6×40
8×40
6×60
12×40
16×40
24×40
method (U-Net: rays × points per ray, by coverage)


M_{12}, U-Net
M_{12}, full CSD
M_{21}, U-Net
M_{21}, full CSD
median |relative error| (%)
0
5
10
15
20
25
30
1
2
5
10
20
50
100
coverage of the grid (%)